

# Tezpur University

## Autumn Mid-Semester Examination–2025

**Course Code:** CS513

**Course Title:** Software Engineering

**Full Marks:** 30

**Time:** 90 Minutes

**Instructions:** There are a total of 7 Questions. Missing data, if any, may be suitably assumed.

1. Consider a new software project that has fixed requirements for resources, and the product needs to be developed within very tight constraints. The project has an estimated 7,35,000 LOC where the project manager has a choice of hiring from two pools of developers using two different cost drivers, such as Analyst Capability (ACAP) and Application Experience (AEXP).

Two cost drivers are considered significant for this project:

| Cost Driver                   | Very Low | Low  | Nominal | High | Very High |
|-------------------------------|----------|------|---------|------|-----------|
| Analyst Capability (ACAP)     | 1.46     | 1.19 | 1.00    | 0.86 | 0.71      |
| Application Experience (AEXP) | 1.29     | 1.13 | 1.00    | 0.91 | 0.82      |

Two possible developer pools are available:

- **Pool 1:** Very highly capable analysts with very little experience in the application domain.
  - **Pool 2:** Analysts of low capability but extensive experience in the application domain.
- a) Apply the Intermediate COCOMO model to compute the effort (in person-months) and development time (in months) if all developers are hired from Pool 1. [2]
  - b) Compute the same for Pool 2. [2]
  - c) Critically evaluate and justify which pool the manager should choose, discussing the quantitative impact (effort/time). [2]
2. Consider a software project with the following information domain characteristics for calculation of Function Point metric:
- Number of output screens and reports = 58
  - Number of prompts and interrupts = 12
  - Number of input screens and tables = 29
  - Number of databases and directories = 07
  - Number of shared databases and shared routines = 04
  - Effort = 36.9 PM

It is given that the complexity weighting factors for EI, EO, EQ, ILF and EIF are 7, 4, 10, 5 and 4 respectively. It is also given that various processing complexity factors are: 3, 4, 1, 2, 4, 2, 5, 0, 3, 3, 0, 2, 5, 4. Then calculate the Function Point (FP) and Productivity (P) for the above software application. [6]

3. A company is developing flight control software for a new drone. The project is highly complex and safety-critical, with partly defined requirements that are expected to evolve. Multiple prototypes and simulations are needed, and continuous risk assessment is essential. Which SDLC model is most suitable by justifying how it supports iterative development, risk management, and evolving requirements. [3]
4. A banking software system is being developed for handling loan approvals and customer accounts. Accuracy, regulatory compliance, and early detection of errors are critical. Each module must be verified and validated against clearly defined requirements before deployment. Identify the SDLC model most appropriate for this project and justify how it ensures thorough verification and validation at each stage. [3]
5. A startup wants to develop a new e-commerce platform, but the requirements are unclear, and the client wants to see a working version quickly for feedback. Frequent changes are expected, and time-to-market is a priority. Which SDLC model would be most suitable? Justify your choice by explaining how it allows rapid prototyping and iterative client feedback. [3]
6. Based on the scenario described below, identify the functional requirements and non-functional requirements (at least 2 from each) of the online food delivery platform.  

A company is developing a comprehensive online food delivery platform aimed at connecting customers with multiple restaurants in a city. The application must allow customers to create and manage accounts, browse restaurants and menus, place orders, make online payments, track delivery status in real-time, and provide ratings and feedback. The system should also support restaurant partners, enabling them to update menu items, manage orders, and track daily sales. Delivery personnel should be able to receive order notifications, update delivery status, and communicate with customers through the platform. In addition to these functionalities, the system must meet several quality expectations. It should be able to handle high traffic during peak hours, maintain data security for customers and payment information, ensure availability and responsiveness 24/7, and provide a user-friendly and intuitive interface for all stakeholders. [2]
7. Draw a complete Use Case Diagram from the following **Online Insurance Policy Management System** scenario showing all actors, use cases, and their relationships.

ABC Insurance Ltd. plans to develop an Online Insurance Policy Management System to streamline its operations and provide a better experience for customers, agents, and administrators. The system is intended to handle all key insurance processes including policy purchase, premium payments, claim submission, and policy management, while ensuring secure and efficient interactions among all stakeholders.

In this system, the primary user is the customer, who can create an account, browse available insurance policies, purchase policies, make premium payments through an integrated payment gateway, and submit claims when necessary. The system guides the customer through these processes, provides notifications for policy renewals, and allows tracking of claim status. Customers interact directly with the system to manage their policies, ensuring convenience and transparency.

Insurance agents act as intermediaries who assist customers in selecting suitable policies, answering queries, and updating policy information as needed. They play a supportive role, ensuring customers make informed decisions and that policy data remains accurate.

The claims officer is responsible for processing claims submitted by customers. They verify the details, assess eligibility, and approve or reject claims according to company policies. This ensures that the claims process is both fair and efficient.

The administrator manages the overall system, including user management, policy configurations, and reporting. They generate operational and financial reports, oversee system security, and ensure smooth functioning of all processes. Additionally, the system interacts with an external payment gateway to securely process online payments. The payment gateway confirms transactions and updates the system regarding payment status, integrating financial operations seamlessly into the workflow. The system is designed to be modular and interactive, with clear relationships between actors and their respective actions. For example, purchasing a policy involves logging in, selecting a policy, processing payment through the gateway, and receiving confirmation. Similarly, submitting a claim involves customer action followed by gateway verification and approval by the claims officer.

[7]